# GAN Exam Answers

**ChatGPT:**

# MODULE 3: GAN Evaluation and Training Objectives

*20 Important Questions – Semester Exam Preparation*

I'll explain all 20 questions in **very simple English**, with easy-to-remember points and real-world examples.

Each answer is organized using:

1. **Definition**
2. **Explanation**
3. **Diagram**
4. **Real-world example**
5. **Advantages**
6. **Disadvantages**
7. **Applications**

For calculation questions, I'll show the formula and step-by-step solution. You can use these answers to prepare for 5-mark and 10-mark questions.

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## 1. Explain the Inception Score (IS) with its formula. How do image quality and diversity affect the score?

### 1. Definition

Inception Score (IS) is an evaluation method used to measure the quality and diversity of images generated by a GAN.

It checks whether generated images are clear and belong to different classes.

## 2. Explanation

IS uses a pretrained Inception Network to classify generated images.

It considers two things:

- **Image quality:** The generated image should clearly belong to a particular class.
- **Image diversity:** The GAN should generate different types of images instead of repeating the same image.

**Formula:**

$$IS = \exp\left(E_x[KL(p(y|x)||p(y))]\right)$$

Where:

- $p(y|x)$ = Probability of a class for a particular image.
- $p(y)$ = Overall probability distribution of classes.
- $KL$ = Kullback-Leibler divergence.
- $\exp$ = Exponential function.

**How it works:**

- Clear images increase the score.
- Different types of images increase the score.
- Repeated images decrease the score.

## 3. Diagram

**Images generated by GAN**

**Pretrained Inception Network**

**Image quality**

*Clear classification*

**Image diversity**

*Different classes*

**Inception Score (IS)**

## 4. Real-world example

Suppose a GAN generates 100 images of animals.

- If all images are clear and contain different animals like dogs, cats and birds, IS will be high.
- If all images are unclear or contain only dogs, IS will be low.

### 5. Advantages

1. Easy to calculate.
2. Measures image quality.
3. Measures image diversity.
4. Uses a pretrained network.

### 6. Disadvantages

1. Does not directly compare generated images with real images.
2. Depends on the Inception Network.
3. May not work well for every type of dataset.
4. A high score does not always mean perfect image quality.

### 7. Applications

1. Evaluating GAN-generated images.
2. Comparing different GAN models.
3. Measuring image generation quality.
4. Research in computer vision.

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## 2. Explain WGAN-GP and the purpose of the gradient penalty.

### 1. Definition

WGAN-GP stands for **Wasserstein Generative Adversarial Network with Gradient Penalty**.

It is an improved GAN model that provides more stable training and reduces problems like mode collapse.

### 2. Explanation

In a normal GAN, the discriminator classifies images as real or fake.

In WGAN-GP, the discriminator is replaced by a **critic**.

The critic gives a score to images instead of classifying them as real or fake.

WGAN-GP uses a gradient penalty to maintain the Lipschitz constraint.

**Gradient penalty:**

It controls the gradients of the critic so that they do not become too large or too small.

**Formula:**

$$L_{GP} = \lambda E[(||\nabla_{\hat{x}} D(\hat{x})||_2 - 1)^2]$$

Where:

- $\lambda$ = Penalty coefficient.
- $\nabla$ = Gradient.
- $\hat{x}$ = Interpolated image.
- $D$ = Critic.

The gradient penalty encourages the gradient norm to be close to 1.

## 3. Diagram

Random Noise

Generator

Generated Images

Critic + Gradient Penalty

**Stable GAN Training**

## 4. Real-world example

Suppose a GAN generates human face images.

During training, the generator may produce distorted faces.

WGAN-GP controls the critic's gradients and helps the generator receive useful feedback to improve the faces.

## 5. Advantages

1. More stable training.
2. Reduces mode collapse.

3. Provides better gradients.
4. Does not require weight clipping.
5. Improves generator learning.

### 6. Disadvantages

1. Requires extra calculations.
2. Training can be slower.
3. Choosing the penalty coefficient is important.

### 7. Applications

1. Face generation.
2. Image generation.
3. Medical image synthesis.
4. Artificial data generation.

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## 3. Calculate FID when real and generated feature means are (2,3) and (4,3), respectively. Both distributions have identity covariance matrices.

### 1. Definition

Fréchet Inception Distance (FID) is a metric used to measure the difference between real and GAN-generated images.

A lower FID value indicates better similarity between the two distributions.

### 2. Explanation

**Given:**

Real image mean:

$$\mu_r = (2, 3)$$

Generated image mean:

$$\mu_g = (4, 3)$$

Both covariance matrices are identity matrices:

$$C_r = C_g = I$$

**FID formula:**

$$FID = ||\mu_r - \mu_g||^2$$
$$+Tr(C_r + C_g - 2(C_r C_g)^{1/2})$$

**Step 1: Calculate the difference between means.**

$$(2-4,\ 3-3) = (-2, 0)$$

**Step 2: Calculate squared distance.**

$$= (-2)^2 + 0^2$$
$$= 4$$

**Step 3: Calculate covariance contribution.**

Since both covariance matrices are identity matrices:

$$Tr(I + I - 2I) = 0$$

**Step 4: Final answer.**

$$FID = 4 + 0$$

*Final Answer: FID = 4*

## 3. Diagram

**Real images**
*Mean (2,3)*

Feature distribution 1

**Generated images**
*Mean (4,3)*

Feature distribution 2

Calculate distance between distributions

**FID = 4**

## 4. Real-world example

Suppose a GAN generates artificial dog images.

FID compares the features of real dog images and generated dog images.

If FID is 4, it indicates a measurable difference between the two distributions. Lower values would generally indicate closer distributions.

### 5. Advantages

1. Compares real and generated images.
2. Measures image similarity.
3. Considers mean and covariance.
4. Useful for comparing GAN models.

### 6. Disadvantages

1. Requires a pretrained network.
2. Calculation can be complex.
3. Results depend on the dataset.
4. Does not measure every aspect of image quality.

### 7. Applications

1. GAN evaluation.
2. Image generation research.
3. Comparing generated and real images.
4. Medical image generation.

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## 4. Explain the difference between a GAN discriminator and a WGAN critic.

### 1. Definition

**GAN Discriminator:** A network that identifies whether an image is real or fake.

**WGAN Critic:** A network that gives a numerical score to measure how realistic an image is.

### 2. Explanation

Both networks provide feedback to the generator, but their working methods are different.

## 3. Diagram

**GAN Discriminator**

Image

Discriminator

Real / Fake

**WGAN Critic**

Image

Critic

Numerical score

## 4. Differences

| GAN Discriminator | WGAN Critic |
|---|---|
| Classifies real or fake. | Gives numerical scores. |
| Uses sigmoid output commonly. | Usually uses linear output. |
| Uses probability. | Uses critic score. |
| Uses cross-entropy loss. | Uses Wasserstein loss. |
| Can suffer from vanishing gradients. | Provides more useful gradients. |
| No Lipschitz constraint in original GAN. | Requires Lipschitz constraint. |
| Helps classify images. | Measures distribution difference. |
| Training can be unstable. | Generally more stable. |

## 5. Real-world example

Suppose a GAN generates a human face.

- Discriminator: Says whether the face is real or fake.
- Critic: Gives a score indicating how the generated face compares with real faces.

## 6. Advantages

**Discriminator:**

- Simple to understand.
- Easy classification.
- Works well in basic GANs.

**Critic:**

- Better gradient feedback.
- More stable training.
- Helps reduce mode collapse.

### 7. Disadvantages

**Discriminator:** May suffer from vanishing gradients.

**Critic:** Requires additional constraints and calculations.

### 8. Applications

- Discriminator: Image generation and image classification-based GAN training.
- Critic: WGAN, WGAN-GP and stable image generation.

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## 5. Explain the objective functions of the generator and discriminator in Min-Max GAN.

### 1. Definition

A Min-Max GAN is a model in which two neural networks, called the generator and discriminator, compete with each other to improve image generation.

### 2. Explanation

There are two main networks.

**1. Generator (G):**

- Takes random noise as input.
- Generates fake images.
- Tries to fool the discriminator.

**2. Discriminator (D):**

- Takes real and fake images.
- Identifies whether images are real or fake.
- Tries to correctly classify images.

**Min-Max objective function:**

$$\min_G \max_D V(D, G)$$

$$V(D, G) = E_{x \sim p_{data}}[\log D(x)]$$

$$+E_{z \sim p_z}[\log(1 - D(G(z)))]$$

Where:

- $G$ = Generator.
- $D$ = Discriminator.
- $x$ = Real image.
- $z$ = Random noise.

**Generator objective:** Fool the discriminator.

**Discriminator objective:** Correctly identify real and fake images.

## 3. Diagram

Random Noise (z)

Generator (G)

Fake Images

Discriminator (D)

Real

Fake

## 4. Real-world example

Suppose the generator creates artificial currency notes.

- Generator creates fake notes.
- Discriminator checks whether they are real or fake.
- Generator improves its notes based on the discriminator's feedback.

This competition continues until the generator produces realistic images.

### 5. Advantages

1. Generates realistic images.
2. Learns automatically.
3. Does not require manually designed output images.
4. Can generate new data.

### 6. Disadvantages

1. Training is difficult.
2. May suffer from mode collapse.
3. Generator and discriminator need proper balance.
4. Training may become unstable.

### 7. Applications

1. Face generation.
2. Image synthesis.
3. Art generation.
4. Data augmentation.
5. Medical image generation.

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## 6. What is mode collapse in GANs? Explain its effect on generated-image diversity and evaluation.

### 1. Definition

Mode collapse is a problem in GAN training where the generator produces only a few types of images repeatedly instead of generating different images.

### 2. Explanation

A GAN should generate different images from random inputs.

But sometimes the generator discovers that producing one particular type of image can fool the discriminator.

As a result, it keeps generating similar images.

**Effects:**

1. Reduces image diversity.
2. Produces repeated images.
3. Limits the generator's learning.
4. Reduces the quality of the dataset.
5. Affects evaluation metrics.

## 3. Diagram

**Normal GAN**



*High diversity*
*Different types of images.*

**Mode collapse**



*Low diversity*
*Repeated similar images.*

## 4. Real-world example

Suppose a GAN is trained to generate different animals.

Expected output:

- Dogs
- Cats
- Birds
- Horses

But due to mode collapse, it generates only dog images repeatedly.

### 5. Advantages

Mode collapse itself has no real advantage because it is a training problem. However, a limited output may sometimes look visually consistent.

### 6. Disadvantages

1. Less diversity.
2. Repeated images.
3. Poor data generation.
4. Reduces model performance.
5. Affects GAN evaluation.

### 7. Applications

Understanding mode collapse is useful in:

1. GAN model evaluation.
2. Image generation.
3. Model debugging.
4. Improving GAN training.
5. Developing stable GAN architectures.

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## 7. Explain the role of a pretrained Inception Network in calculating IS and FID.

### 1. Definition

A pretrained Inception Network is a deep learning model already trained on a large image dataset to recognize objects and extract image features.

It is used to evaluate the quality of GAN-generated images.

### 2. Explanation

The Inception Network plays an important role in calculating two metrics.

**1. Role in Inception Score (IS):**

- Takes generated images as input.
- Classifies the images into different categories.
- Measures prediction confidence and class diversity.
- Calculates the IS value.

**2. Role in FID:**

- Extracts features from real images.
- Extracts features from generated images.
- Calculates feature means and covariance matrices.
- Compares both distributions.

## 3. Diagram

Real Images

Generated Images

Pretrained Inception Network

IS

FID

## 4. Real-world example

Suppose a GAN generates images of cars.

The Inception Network identifies features such as shape, wheels and other visual patterns.

It helps measure whether the generated cars are recognizable and whether their features are similar to real car images.

## 5. Advantages

1. Already trained model.
2. Saves training time.
3. Extracts useful image features.
4. Helps calculate IS and FID.

### 6. Disadvantages

1. May not understand specialized images.
2. Depends on training data.
3. May not capture every visual detail.
4. Requires computational resources.

### 7. Applications

1. GAN evaluation.
2. Image classification.
3. Feature extraction.
4. Image quality measurement.

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## 8. For a real sample, D(x) = 0.8, and for a generated sample, D(G(z)) = 0.25. Calculate the discriminator log-likelihood contribution using natural logarithms.

### 1. Definition

Discriminator log-likelihood measures how well the discriminator identifies real and generated images.

### 2. Explanation

**Given:**

$$D(x) = 0.8$$
$$D(G(z)) = 0.25$$

**Formula:**

$$L_D = \log D(x) + \log(1 - D(G(z)))$$

**Step 1: Substitute values.**

$$L_D = \ln(0.8) + \ln(1 - 0.25)$$

**Step 2: Simplify.**

$$= \ln(0.8) + \ln(0.75)$$

**Step 3: Calculate natural logarithms.**

$$\ln(0.8) = -0.2231$$

$$\ln(0.75) = -0.2877$$

**Step 4: Add the values.**

$$L_D = -0.2231 - 0.2877$$

*Final Answer = -0.5108*

## 3. Diagram

**Real sample**

*D(x)*

**0.8**

**Generated sample**

*D(G(z))*

**0.25**

---

**Log-likelihood = -0.5108**

## 4. Real-world example

Suppose a discriminator checks real and fake currency notes.

- It gives 0.8 probability that a real note is real.
- It gives 0.25 probability that a fake note is real.

Therefore, it correctly assigns 0.75 probability to the fake note.

## 5. Advantages

1. Measures discriminator performance.
2. Helps understand GAN training.
3. Uses a mathematical objective.
4. Helps compare predictions.

## 6. Disadvantages

1. Logarithm calculations may be difficult.

2. Values can be negative.
3. Does not directly measure image quality.

### 7. Applications

1. GAN training.
2. Discriminator evaluation.
3. Loss calculation.
4. Deep learning optimization.

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## 9. Explain the training objectives of the original Min-Max GAN and the non-saturating GAN. Why is the non-saturating objective often preferred?

### 1. Definition

Min-Max GAN and non-saturating GAN are two different training objectives used to train the generator and discriminator.

### 2. Explanation

**1. Original Min-Max GAN:**

The generator tries to minimize the discriminator's ability to identify fake images.

Formula:

$$L_G = E_z[\log(1 - D(G(z)))]$$

**2. Non-saturating GAN:**

The generator directly tries to increase the discriminator's prediction that generated images are real.

Formula:

$$L_G = -E_z[\log D(G(z))]$$

**Why is non-saturating GAN preferred?**

- It provides stronger gradients when the discriminator is confident that fake images are fake.
- It helps the generator learn faster.
- It reduces the vanishing-gradient problem during early training.

## 3. Diagram

**Min-Max GAN**

Generator minimizes:
$\log(1 - D(G(z)))$

**Non-saturating GAN**

Generator minimizes:
$-\log D(G(z))$

Generator improves generated images

## 4. Real-world example

Suppose a GAN generates human faces.

Initially, the discriminator easily identifies fake faces.

- Original Min-Max GAN may provide weak learning signals.
- Non-saturating GAN provides stronger gradients, helping the generator improve faster.

## 5. Advantages

**Min-Max GAN:**

1. Follows the original GAN objective.
2. Has a clear mathematical foundation.

**Non-saturating GAN:**

1. Stronger gradients.
2. Faster generator learning.
3. Reduces vanishing gradients.

## 6. Disadvantages

**Min-Max GAN:**

1. Generator may learn slowly.
2. Can suffer from vanishing gradients.

**Non-saturating GAN:**

1. Does not remove all GAN training problems.
2. Training can still be unstable.

### 7. Applications

1. Image generation.
2. Face generation.
3. Deep learning research.
4. GAN optimization.

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## 10. Describe the Lipschitz constraint in WGAN and explain why it is necessary.

### 1. Definition

The Lipschitz constraint is a mathematical condition that controls how much the output of the WGAN critic can change when its input changes.

In WGAN, the critic must be 1-Lipschitz.

### 2. Explanation

A function is 1-Lipschitz if:

$$|D(x_1) - D(x_2)| \leq ||x_1 - x_2||$$

It means the change in the critic's output should not be greater than the change in its input.

**Why is it necessary?**

1. Maintains stable critic behaviour.
2. Helps calculate Wasserstein distance.
3. Prevents extremely large gradients.
4. Provides useful feedback to the generator.
5. Improves training stability.

Common methods:

- Weight clipping.
- Gradient penalty (WGAN-GP).

- Spectral normalization.

## 3. Diagram

Real and generated images

WGAN Critic

Apply 1-Lipschitz constraint

**Stable gradients and training**

## 4. Real-world example

Suppose a WGAN evaluates generated human faces.

A small change in the input image should not cause an extremely large change in the critic's score.

The Lipschitz constraint controls this change and helps maintain stable learning.

## 5. Advantages

1. Improves training stability.
2. Controls gradients.
3. Helps calculate Wasserstein distance.
4. Provides better feedback.

## 6. Disadvantages

1. Requires additional calculations.
2. Applying the constraint can be difficult.
3. May increase training time.

## 7. Applications

1. WGAN.
2. WGAN-GP.
3. Image generation.
4. Stable GAN training.

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# 11. Explain Fréchet Inception Distance (FID), including the role of feature means and covariance matrices.

## 1. Definition

Fréchet Inception Distance (FID) is an evaluation metric used to measure the similarity between real images and GAN-generated images.

A **lower FID score indicates better image quality and similarity**.

## 2. Explanation

FID uses a pretrained Inception Network to extract features from real and generated images.

It calculates two important values:

**1. Feature Mean:**

- Represents the average features of images.
- Measures the overall difference between real and generated image features.

**2. Covariance Matrix:**

- Represents the variation and relationship between image features.
- Helps measure differences in image diversity and distribution.

**FID Formula:**

$$FID = ||\mu_r - \mu_g||^2$$
$$+Tr(C_r + C_g - 2(C_rC_g)^{1/2})$$

Where:

- $\mu_r$ = Mean of real image features.
- $\mu_g$ = Mean of generated image features.
- $C_r$ = Real image covariance matrix.
- $C_g$ = Generated image covariance matrix.

## 3. Diagram

Real Images

Generated Images

Inception Network

Feature Mean

Covariance

**Calculate FID Score**

## 4. Real-world example

Suppose a GAN generates artificial human faces.

FID compares the features of real human faces with generated faces.

- FID = 5: Relatively close distributions.
- FID = 50: Much larger difference.

The exact quality interpretation depends on the dataset and evaluation setup.

## 5. Advantages

1. Compares real and generated images.
2. Measures feature similarity.
3. Considers image diversity.
4. Useful for comparing GAN models.

## 6. Disadvantages

1. Requires a pretrained network.
2. Computationally expensive.
3. Depends on dataset size.
4. Does not perfectly measure human visual quality.

## 7. Applications

1. Image generation.
2. Face synthesis.
3. Medical image generation.
4. GAN performance evaluation.

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# 12. Explain the architecture and working principle of Wasserstein GAN (WGAN). What is the role of the critic and the Lipschitz constraint?

## 1. Definition

Wasserstein GAN (WGAN) is an improved GAN architecture that uses Wasserstein distance to measure the difference between real and generated image distributions.

It helps make GAN training more stable.

## 2. Explanation

WGAN contains two main networks.

**1. Generator:**

- Takes random noise as input.
- Generates fake images.
- Tries to produce realistic images.

**2. Critic:**

- Receives real and generated images.
- Assigns scores to images.
- Helps measure the difference between both distributions.

**Working principle:**

1. Random noise is given to the generator.
2. Generator produces fake images.
3. Critic evaluates real and fake images.
4. The difference between average scores is calculated.
5. Generator improves based on critic feedback.
6. This process repeats until generated images improve.

**Wasserstein distance:**

$$W(P_r, P_g) = \sup_{||f||_L \leq 1} (E[f(x)] - E[f(G(z))])$$

The Lipschitz constraint ensures the critic satisfies the required mathematical condition.

## 3. Diagram

Random Noise

Generator

Generated Images

Critic + Lipschitz Constraint

Wasserstein Distance

## 4. Real-world example

Suppose WGAN generates artificial medical images.

The critic compares real medical images with generated images and gives scores.

The generator uses this feedback to improve the generated images.

## 5. Advantages

1. Stable training.
2. Reduces mode collapse.
3. Better gradient flow.
4. Provides meaningful training feedback.
5. Improves image generation.

## 6. Disadvantages

1. Requires a Lipschitz constraint.
2. Training can be computationally expensive.
3. Requires careful parameter selection.

## 7. Applications

1. Medical image generation.
2. Face generation.
3. Artificial data generation.
4. Image synthesis.
5. Computer vision.

# 13. Discuss practical signs that may indicate a GAN is ready to stop training.

## 1. Definition

Stopping criteria are conditions used to decide when GAN training should be stopped because the model has learned to generate sufficiently good images.

## 2. Explanation

A GAN should not be trained forever.

We can stop training when the following conditions are observed:

1. **Good image quality:** Generated images look realistic.
2. **High diversity:** Generator produces different types of images.
3. **Stable loss:** Generator and discriminator losses are relatively stable.
4. **Good FID score:** FID becomes low and stops improving significantly.
5. **No mode collapse:** Images are not repeated.
6. **Validation performance:** Generated images maintain good quality on evaluation data.

Note: GAN losses do not always decrease smoothly, so loss alone should not decide when to stop training.

## 3. Diagram

Start GAN Training

Evaluate Images, Loss and FID

Good Quality and Diversity?

**Yes**

Stop Training

**No**

Continue Training

## 4. Real-world example

Suppose a GAN generates 10,000 artificial human faces.

Initially, faces may look distorted.

After training, faces become realistic, diverse and the FID score stops improving significantly.

At this stage, we can stop training and save the best model checkpoint.

### 5. Advantages

1. Saves training time.
2. Reduces unnecessary computation.
3. Prevents unnecessary overtraining.
4. Helps select the best model.

### 6. Disadvantages

1. Choosing stopping conditions is difficult.
2. GAN losses may fluctuate.
3. Low FID alone does not guarantee perfect quality.

### 7. Applications

1. GAN model development.
2. Image generation.
3. Deep learning research.
4. Model optimization.

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## 14. Explain the non-saturating generator loss and how it can improve generator learning.

### 1. Definition

Non-saturating generator loss is a training objective used in GANs to help the generator learn faster by providing stronger gradients.

### 2. Explanation

In a normal Min-Max GAN, the generator tries to minimize:

$$L_G = E_z[\log(1 - D(G(z)))]$$

Sometimes the discriminator easily identifies fake images, causing weak gradients for the generator.

To solve this problem, the non-saturating loss is used.

**Formula:**

$$L_G = -E_z[\log D(G(z))]$$

**How it works:**

1. Generator creates fake images.
2. Discriminator checks the images.
3. Generator receives feedback.
4. It tries to increase the probability that fake images are classified as real.
5. It improves the generated images.

## 3. Diagram

Random Noise

Generator

Fake Images

Discriminator Feedback

Stronger Generator Learning

## 4. Real-world example

Suppose a GAN generates human faces.

Initially, the discriminator identifies fake faces easily.

The non-saturating loss gives stronger learning signals to the generator so it can improve facial features such as eyes, nose and mouth.

## 5. Advantages

1. Provides stronger gradients.
2. Improves generator learning.
3. Reduces the vanishing-gradient problem.
4. Helps faster learning.

### 6. Disadvantages

1. Does not solve all GAN training problems.
2. Training can still be unstable.
3. Requires proper tuning.

### 7. Applications

1. Face generation.
2. Image synthesis.
3. GAN training.
4. Artificial image generation.

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## 15. A WGAN critic gives real and generated samples average scores of 0.9 and 0.4. Calculate the Wasserstein gap.

### 1. Definition

Wasserstein gap is the difference between the average critic scores of real images and generated images.

### 2. Explanation

**Given:**

Average real score:

$$E[D(x)] = 0.9$$

Average generated score:

$$E[D(G(z))] = 0.4$$

**Formula:**

$$Wasserstein\ Gap = E[D(x)] - E[D(G(z))]$$

**Step 1: Substitute values.**

$$= 0.9 - 0.4$$

**Step 2: Calculate.**

$$= 0.5$$

*Final Answer = 0.5*

**Interpretation:**

The Wasserstein gap is 0.5. It indicates a difference between real and generated image scores.

During training, a decreasing gap can indicate that the distributions are becoming closer, provided the critic is trained properly.

## 3. Diagram

**Real images**

0.9

**Generated images**

0.4

0.9 - 0.4 = 0.5

## 4. Real-world example

Suppose a WGAN generates artificial car images.

The critic gives:

- Real cars: 0.9
- Generated cars: 0.4

The difference is 0.5, which is the Wasserstein gap.

## 5. Advantages

1. Easy to calculate.
2. Helps monitor GAN training.
3. Measures the difference in critic scores.
4. Useful for evaluating WGAN training progress.

## 6. Disadvantages

1. Depends on critic quality.
2. Does not directly measure visual quality.

3. Requires a proper Lipschitz constraint.

### 7. Applications

1. WGAN training.
2. GAN evaluation.
3. Image generation.
4. Monitoring model performance.

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## 16. Explain how increasing generator and discriminator network depth affects feature learning, gradient flow and GAN training stability.

### 1. Definition

Network depth refers to the number of layers present in a neural network.

Increasing the depth of the generator and discriminator can improve feature learning, but it may also make GAN training difficult.

### 2. Explanation

**1. Effect on feature learning:**

- More layers learn complex features.
- Generator can create detailed images.
- Discriminator can identify small differences.

**2. Effect on gradient flow:**

- Deep networks may suffer from vanishing gradients.
- Sometimes gradients become too large, causing exploding gradients.
- This makes learning difficult.

**3. Effect on training stability:**

- Very deep networks may become difficult to balance.
- Generator and discriminator may learn at different speeds.
- Training may become unstable.

**Methods to improve stability:**

1. Batch normalization.
2. Residual connections.
3. Proper weight initialization.
4. Gradient penalty.
5. Suitable learning rates.

## 3. Diagram

Input Image / Noise

Layer 1 → Layer 2 → Layer 3

Layer 4 → Layer 5 → Layer 6

Complex Feature Learning

*Increasing layers can learn more complex features but may create gradient problems.*

## 4. Real-world example

Suppose a GAN generates human faces.

- A shallow generator may generate simple face shapes.
- A deeper generator can learn detailed eyes, hair and skin texture.
- If it becomes too deep, gradients may become unstable and training may fail.

## 5. Advantages

1. Learns complex features.
2. Generates detailed images.
3. Improves representation power.
4. Can improve image quality.

## 6. Disadvantages

1. Requires more memory.
2. Needs more training time.
3. Can suffer from vanishing gradients.
4. Training becomes difficult.
5. May become unstable.

### 7. Applications

1. High-resolution image generation.
2. Face synthesis.
3. Medical image generation.
4. Deep learning and computer vision.

---

## 17. Explain the Inception Score (IS) and describe one limitation of this evaluation metric.

### 1. Definition

Inception Score (IS) is a metric used to evaluate the quality and diversity of images generated by GANs.

A higher IS generally indicates better image quality and diversity.

### 2. Explanation

IS uses a pretrained Inception Network.

It checks:

1. Whether an image belongs clearly to a particular class.
2. Whether the generated images contain different classes.

**Formula:**

$$IS = \exp(E_x[KL(p(y|x)||p(y))])$$

Where:

- $p(y|x)$ = Class probability for an image.
- $p(y)$ = Overall class probability.
- KL = Kullback-Leibler divergence.

**One important limitation:**

IS does not directly compare generated images with real images.

Therefore, it may give a high score even when generated images are different from the real dataset.

## 3. Diagram

GAN-generated images

Inception Network

Quality

Diversity

**Inception Score**

## 4. Real-world example

Suppose a GAN generates 500 animal images.

If the images are clear and contain dogs, cats, birds and horses, the IS may be high.

However, the images may still look different from real animal photographs.

## 5. Advantages

1. Easy to calculate.
2. Measures image quality.
3. Considers diversity.
4. Useful for GAN comparison.

## 6. Disadvantages

1. Does not compare with real images.
2. Depends on the Inception Network.
3. May not detect repeated images within a class.
4. Not suitable for every dataset.

## 7. Applications

1. GAN evaluation.
2. Image synthesis.
3. Computer vision research.
4. Comparing GAN models.

---

# 18. Explain why IS and FID should be interpreted carefully rather than used as the only measures of GAN quality.

## 1. Definition

IS and FID are evaluation metrics used to measure the quality of GAN-generated images, but they cannot completely measure all aspects of image quality.

## 2. Explanation

Both metrics have different advantages and limitations.

**Inception Score (IS):**

- Measures image quality and diversity.
- Does not directly compare generated images with real images.

**Fréchet Inception Distance (FID):**

- Compares real and generated image feature distributions.
- Uses feature means and covariance matrices.
- Lower FID generally indicates better similarity.

**Why should we interpret them carefully?**

1. Metrics may not match human opinions.
2. IS cannot directly compare real and generated distributions.
3. FID depends on the Inception Network.
4. Dataset size can affect results.
5. Both metrics may miss some visual defects.

Therefore, we should also use human visual inspection and other evaluation methods.

## 3. Diagram

GAN-generated Images

IS

FID

Human Visual Inspection

**Overall GAN Quality Evaluation**

### 4. Real-world example

Suppose two GAN models generate human faces.

- Model A has a good FID score but produces some strange-looking eyes.
- Model B has a slightly higher FID score but produces faces that look more natural to humans.

Therefore, we should not select a model using only one metric.

### 5. Advantages

1. IS and FID provide numerical evaluation.
2. Help compare different models.
3. Reduce manual evaluation effort.
4. Help monitor training.

### 6. Disadvantages

1. Cannot completely measure human visual quality.
2. Results depend on datasets.
3. May miss small defects.
4. Require additional evaluation methods.

### 7. Applications

1. GAN performance evaluation.
2. Image generation research.
3. Model comparison.
4. Selecting the best generator.

---

## 19. Why can increasing the depth of a GAN generator or discriminator make training more difficult? Discuss gradient flow and stability.

### 1. Definition

Increasing network depth means adding more layers to the generator and discriminator.

Although more layers help learn complex features, they can make GAN training difficult because of gradient flow and stability problems.

## 2. Explanation

**1. Gradient flow problem:**

During training, gradients travel backward through network layers.

Two major problems can occur:

- **Vanishing gradient:** Gradients become very small, so earlier layers learn slowly.
- **Exploding gradient:** Gradients become very large, making training unstable.

**2. Training stability:**

- Deep networks require more computational power.
- Generator and discriminator may become unbalanced.
- Training may become unstable.
- Model convergence becomes difficult.

**Solutions:**

1. Batch normalization.
2. Residual connections.
3. Gradient penalty.
4. Proper learning rate.
5. Weight initialization.

## 3. Diagram

Input

Layer 1 → Layer 2

Layer 3 → Layer 4

Layer 5 → Layer 6

Vanishing / Exploding Gradients

Training Instability

## 4. Real-world example

Suppose we build a GAN to generate realistic human faces.

A deeper generator can learn details such as eyes, nose, hair and skin.

But if the network has too many layers, gradients may become weak and the generator may fail to learn these features properly.

### 5. Advantages

1. Learns complex features.
2. Generates detailed images.
3. Improves feature representation.
4. Can improve image quality.

### 6. Disadvantages

1. Vanishing gradients.
2. Exploding gradients.
3. Requires more memory.
4. Training becomes slower.
5. Can cause instability.

### 7. Applications

1. Face generation.
2. Image synthesis.
3. Medical image processing.
4. Computer vision.
5. Deep learning.

---

## 20. Explain how the quality and diversity of generated images can affect the evaluation of a GAN.

### 1. Definition

Image quality and diversity are two important factors used to evaluate the performance of a Generative Adversarial Network (GAN).

- **Image quality:** How realistic and clear the generated images are.
- **Image diversity:** How many different types of images the generator can produce.

## 2. Explanation

Both factors are important for evaluating GAN performance.

**1. Effect of image quality:**

- Clear images indicate good generation.
- Realistic images indicate better learning.
- Blurred or distorted images indicate poor generation.

**2. Effect of image diversity:**

- Different images indicate good diversity.
- Repeated images indicate low diversity.
- Mode collapse reduces diversity.

**Evaluation metrics:**

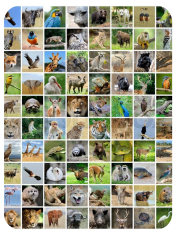
- IS (Inception Score): Measures image quality and diversity.
- FID (Fréchet Inception Distance): Compares real and generated image distributions.

**Important point:**

A good GAN should generate images that are both realistic and diverse.

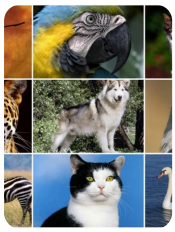
## 3. Diagram

**GAN-generated images**



**High quality**

*Clear images*



**High diversity**

*Different images*

**Better GAN Evaluation**

## 4. Real-world example

Suppose a GAN generates 1,000 animal images.

**Case 1: High quality but low diversity**

- All images are clear.
- All images contain dogs.
- The generator lacks diversity.

**Case 2: Low quality but high diversity**

- Images contain dogs, cats, birds and horses.
- But images are blurred and distorted.

**Case 3: High quality and high diversity**

- Images are clear and realistic.
- Different animals are generated.

Case 3 indicates better GAN performance.

## 5. Advantages

1. Helps evaluate GAN performance.
2. Identifies image generation problems.
3. Helps improve image quality.
4. Detects mode collapse.
5. Helps compare different GAN models.

## 6. Disadvantages

1. Evaluation can be computationally expensive.
2. IS and FID may not match human opinions.
3. Measuring diversity can be difficult.
4. Requires proper evaluation datasets.

### 7. Applications

1. Face generation.
2. Medical image generation.
3. Artificial data generation.
4. Image synthesis.
5. Computer vision research.

---

# Important formulas – Quick revision

These are the most important formulas from your Module 3. Revise them before the examination.

| Topic | Formula |
|---|---|
| Inception Score | $IS=\exp(E[KL(p(y\|x) |
| FID | $ |
| Min-Max GAN | $\min_G \max_D V(D,G)$ |
| Original Generator Loss | $E[\log(1 - D(G(z)))]$ |
| Non-saturating Loss | $-E[\log D(G(z))]$ |
| Gradient Penalty | $\lambda E[( |
| Wasserstein Gap | $E[D(x)] - E[D(G(z))]$ |
| Lipschitz Constraint | $ |

## Important differences to remember

| Topic | Main difference |
|---|---|
| IS vs FID | IS measures quality and diversity; FID compares real and generated distributions. |
| GAN vs WGAN | GAN uses a discriminator; WGAN uses a critic. |
| Discriminator vs Critic | Discriminator gives real/fake probability; critic gives scores. |

| Topic | Main difference |
|---|---|
| Min-Max vs Non-saturating | Non-saturating loss provides stronger gradients to the generator. |
| Normal GAN vs Mode Collapse | Normal GAN generates diverse images; mode collapse generates repeated images. |

## Most important questions for exam revision

If you have less time, prioritize these topics:

*High priority*

1. Q1 – Inception Score (IS)
2. Q2 – WGAN-GP
3. Q3 – FID numerical problem
4. Q5 – Min-Max GAN
5. Q6 – Mode Collapse
6. Q9 – Min-Max vs Non-saturating GAN
7. Q10 – Lipschitz Constraint
8. Q11 – FID
9. Q12 – WGAN Architecture
10. Q15 – Wasserstein Gap numerical problem

**My suggestion:** First prepare Q1–Q15, especially the formulas and numerical problems. Then revise Q16–Q20 because they cover related concepts such as network depth, gradient flow, quality and diversity.

You don't need to memorize every sentence. Understand the definition, 4–5 explanation points, diagram and example for each answer. That will make it much easier to write in your semester.

---

**You:**

Same as module 3 give for module 4

MODULE 4: Semi-Supervised GAN and Conditional GAN

1. Explain the motivation for Conditional GANs and how conditioning controls the generated output.

2. Explain the training process of a Semi-Supervised GAN using labeled real data, unlabeled real data and generated samples.

3. Draw and explain the architecture of a CGAN generator. How are the noise vector and condition combined?

4. Differentiate between supervised loss and unsupervised loss in a Semi-Supervised GAN.

5. A Semi-Supervised GAN correctly classifies 40 out of 50 labeled images. Calculate the classification accuracy. If 30 out of 50 unlabeled real images are classified as real rather than fake, calculate this proportion.

6. Explain the working of a CGAN discriminator and identify the inputs it receives during training.

7. Explain two benefits and two limitations of Semi-Supervised GANs.

8. Explain the architecture of a CGAN discriminator. How does it evaluate an image together with its condition or class label?

9. Describe the complete training procedure of a Conditional GAN, from input preparation to generator and discriminator updates.

10. Explain the role of labeled, unlabeled and generated samples in Semi-Supervised GAN training.

11. Explain how class labels can guide image generation in a CGAN, using a suitable example.

12. Compare a Semi-Supervised GAN with a fully supervised classifier in terms of labeled data, architecture, training objectives and limitations. 13. Outline the major implementation steps required to build and train a CGAN.

14. Describe the architecture of a Semi-Supervised GAN and explain the purpose of the additional (K+1)th class.

15. What problems may occur when the condition supplied to a CGAN is incorrect or inconsistent with the training image?

16. Explain the training objectives of a Semi-Supervised GAN. Differentiate between supervised classification loss and unsupervised loss.

17. Compare an unconditional GAN with a Conditional GAN in terms of input, control over output and applications.

18. Explain how a Semi-Supervised GAN can learn from unlabeled real images in addition to labeled images.

19. Explain the implementation steps of a Semi-Supervised GAN and discuss any two challenges encountered during training.

20. Explain how generated samples contribute to the training of a Semi-Supervised GAN.

---

**You:**

Give me the content for now..

---

**ChatGPT:**

# MODULE 4: Semi-Supervised GAN and Conditional GAN

*Semester Exam Preparation 20 Important Questions*

These answers are written in **very simple English**, with real-world examples and easy diagrams. They are suitable for writing in your semester examination.

Each answer follows this format:

1. Definition
2. Explanation
3. Diagram
4. Real-world example
5. Advantages
6. Disadvantages
7. Applications

For numerical questions, formulas and step-by-step calculations are included.

---

# PART A: CONDITIONAL GAN (CGAN)

## Q1. Explain the motivation for Conditional GANs and how conditioning controls the generated output.

### 1. Definition

A Conditional Generative Adversarial Network (CGAN) is a type of GAN that generates images based on additional information called a **condition**.

The condition may be a class label, text description or another image.

### 2. Explanation

- A normal GAN generates images using only random noise.
- It cannot directly control which type of image should be generated.
- CGAN solves this problem by providing extra information to the generator.
- The generator receives random noise and a condition.
- The condition guides the generator to produce the required output.
- The discriminator checks whether the generated image is real and matches the given condition.

**How conditioning works:**

- Condition = Cat → Generates a cat image.
- Condition = Dog → Generates a dog image.
- Condition = Car → Generates a car image.

### 3. Diagram

```
Random Noise (z)
      +
Condition (y)
      |
      v
  Generator
      |
      v
Generated Image
      |
      v
 Discriminator
      |
```

```
      v
Real or Fake?
```

### 4. Real-world example

Suppose we want to generate handwritten digits.

If we provide the condition 5, the CGAN generates an image of digit 5. If we provide 8, it generates digit 8.

### 5. Advantages

1. Provides control over generated images.
2. Generates specific classes of images.
3. Supports text-based image generation.
4. Produces different images for the same condition.

### 6. Disadvantages

1. Requires correct condition labels.
2. Training is more complex than a normal GAN.
3. Incorrect conditions may produce wrong images.

### 7. Applications

- Image generation.
- Face generation.
- Text-to-image generation.
- Medical image generation.
- Image-to-image translation.

---

## Q2. Explain the training process of a Semi-Supervised GAN using labeled real data, unlabeled real data and generated samples.

### 1. Definition

A Semi-Supervised GAN is a GAN that uses **labeled real data, unlabeled real data and generated fake data** to train a model.

It helps the model learn even when only a small amount of labeled data is available.

## 2. Explanation

Semi-Supervised GAN uses three types of data.

### 1. Labeled real data

- These images have class labels.
- They train the model to identify the correct class.
- Example: Images labeled as cats, dogs and birds.

### 2. Unlabeled real data

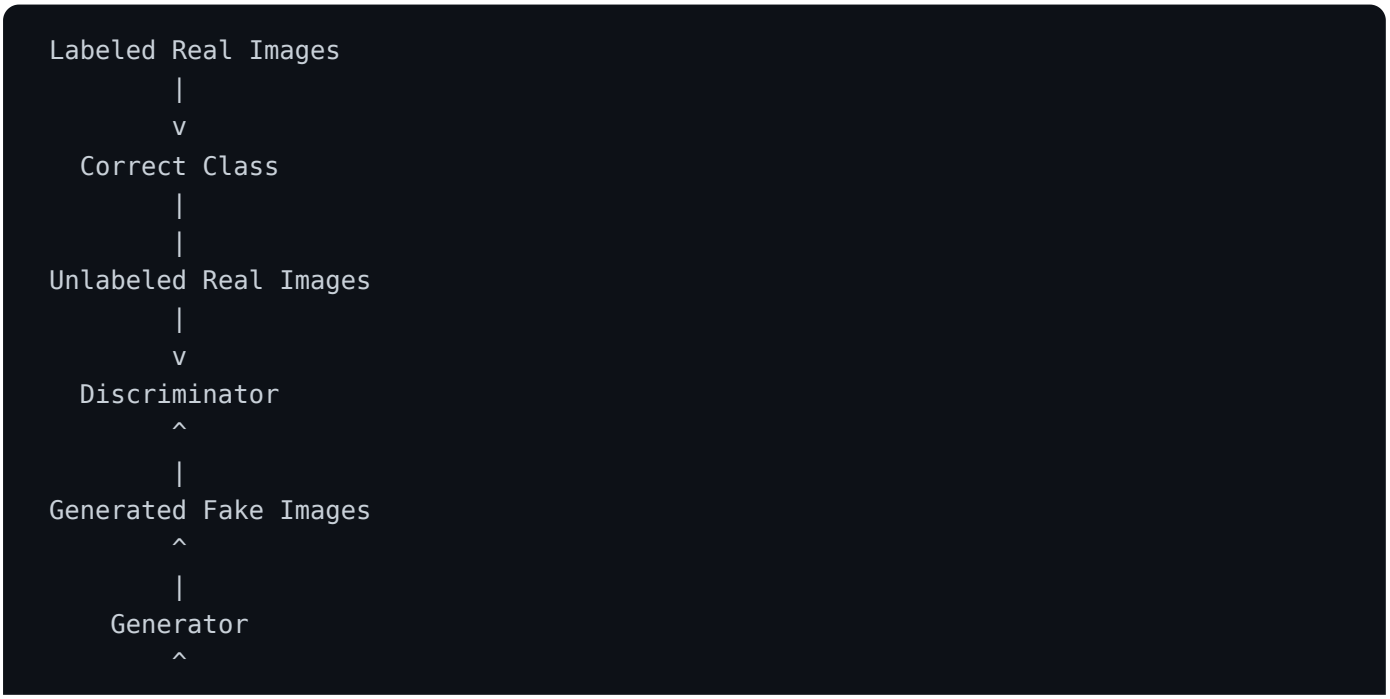
- These are real images without class labels.
- The model learns that these images belong to real data.
- It uses them to improve its understanding of the dataset.

### 3. Generated fake data

- The generator creates fake images using random noise.
- The discriminator learns to identify these images as fake.
- The generator tries to produce more realistic images.

The discriminator has K+1 output classes, where K represents real classes and one additional class represents fake images.

## 3. Diagram



```
          |
     Random Noise

Output: K Real Classes + 1 Fake Class
```

## 4. Real-world example

Suppose a hospital has 10,000 medical images.

- 1,000 images have labels.
- 9,000 images have no labels.
- The generator creates additional fake images.

The model learns from all these images instead of depending only on the 1,000 labeled images.

## 5. Advantages

1. Reduces the need for labeled data.
2. Uses unlabeled data effectively.
3. Improves classification learning.
4. Can learn useful image features.

## 6. Disadvantages

1. Training is complex.
2. Generator and discriminator may become unbalanced.
3. Poor generated images may affect learning.

## 7. Applications

- Medical image classification.
- Face recognition.
- Object detection.
- Image classification.

---

# Q3. Draw and explain the architecture of a CGAN generator. How are the noise vector and condition combined?

## 1. Definition

A CGAN generator is a neural network that receives a random noise vector and a condition to generate an image belonging to the required class.

## 2. Explanation

The generator works using the following steps:

1. **Random Noise (z):** Provides randomness and variation.
2. **Condition (y):** Provides information about the required output.
3. **Concatenation:** Noise and condition are combined into a single vector.
4. **Generator Network:** The combined vector passes through neural network layers.
5. **Output:** The generator produces an image matching the given condition.

The most common method of combining noise and condition is called **concatenation**.

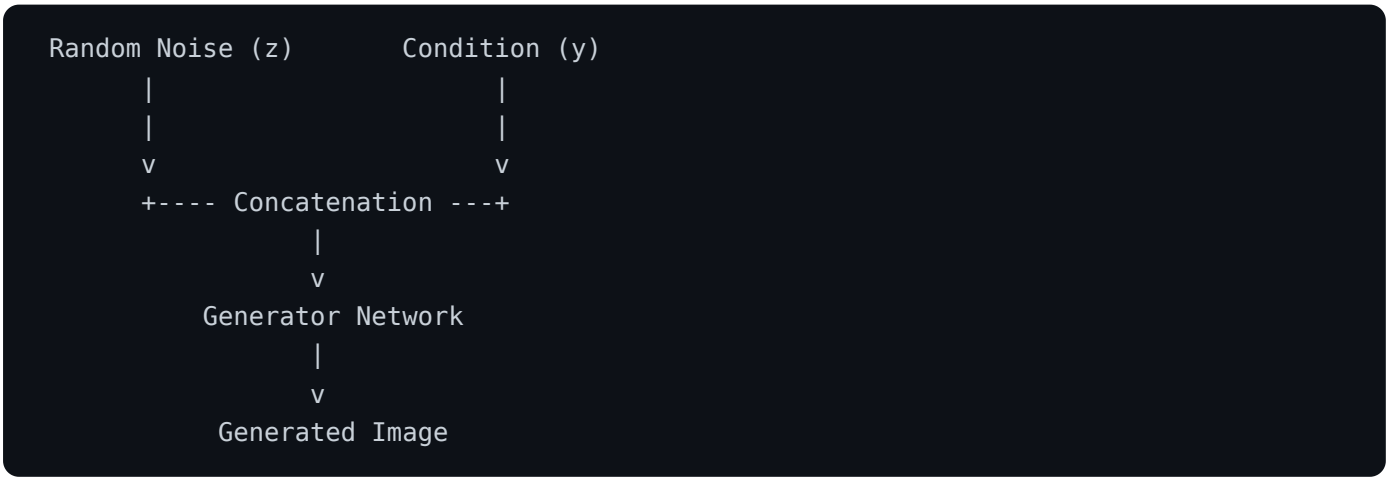
Formula:

$$G(z, y) = \text{Generated Image}$$

Where:

- z = Random noise.
- y = Condition or class label.
- G = Generator.

## 3. Diagram



### 4. Real-world example

In handwritten digit generation:

- Random noise provides different handwriting styles.
- Condition 7 tells the generator to create digit 7.
- The generator produces different handwritten versions of 7.

### 5. Advantages

1. Controls the generated output.
2. Produces different images for the same class.
3. Simple method of combining inputs.

### 6. Disadvantages

1. Requires correct condition information.
2. Generator may ignore the condition.
3. Training can be difficult.

### 7. Applications

- Handwritten digit generation.
- Face generation.
- Image synthesis.
- Fashion image generation.

---

## Q4. Differentiate between supervised loss and unsupervised loss in a Semi-Supervised GAN.

### 1. Definition

**Supervised loss:** It measures the error between the predicted class and the actual class of labeled real images.

**Unsupervised loss:** It helps the model distinguish between unlabeled real images and generated fake images.

## 2. Explanation

### Supervised Loss

- Uses labeled real images.
- Compares predicted labels with actual labels.
- Helps classify images into correct categories.
- Example: Identifying an image as a cat or dog.

Formula:

$$L_{sup} = -E_{(x,y)}[\log p(y|x)]$$

### Unsupervised Loss

- Uses unlabeled real images and generated fake images.
- Real images should not be classified as fake.
- Generated images should be classified as fake.
- Helps the model distinguish real and fake images.

Formula:

$$L_{unsup} = -E_x[\log(1 - p(fake|x))]$$
$$-E_z[\log p(fake|G(z))]$$

## 3. Diagram

```
Labeled Real Images
        |
        v
  Supervised Loss
        |
        v
   Correct Class

Unlabeled Real Images
        +
Generated Fake Images
        |
        v
 Unsupervised Loss
        |
        v
  Real / Fake
```

### 4. Real-world example

A flower classification system uses:

- Labeled rose and lily images for supervised learning.
- Unlabeled flower images and generated fake images for unsupervised learning.

### 5. Advantages

1. Supervised loss provides correct class information.
2. Unsupervised loss uses unlabeled data.
3. Both losses improve model learning.

### 6. Disadvantages

1. Supervised loss requires labeled images.
2. Unsupervised loss may be affected by poor generated images.
3. Balancing both losses is difficult.

### 7. Applications

- Image classification.
- Medical image analysis.
- Object recognition.
- Semi-supervised learning.

---

## Q5. A Semi-Supervised GAN correctly classifies 40 out of 50 labeled images. Calculate the classification accuracy. If 30 out of 50 unlabeled real images are classified as real rather than fake, calculate this proportion.

### 1. Definition

Classification accuracy is the percentage of correctly classified images out of the total number of images.

Proportion represents the fraction of images classified as real out of the total unlabeled real images.

## 2. Explanation and calculation

**Part 1: Classification accuracy**

Given:

- Correctly classified images = 40
- Total labeled images = 50

Formula:

$$Accuracy = \frac{Correct}{Total} \times 100$$

Substitution:

$$= \frac{40}{50} \times 100$$

$$= \boxed{80\%}$$

**Part 2: Proportion classified as real**

Given:

- Images classified as real = 30
- Total unlabeled real images = 50

Formula:

$$Proportion = \frac{Real\ images}{Total\ images}$$

Substitution:

$$= \frac{30}{50} = 0.6$$

$$= \boxed{60\%}$$

## 3. Diagram

```
Labeled Images: 50
       |
       v
Correct = 40
       |
       v
Accuracy = 80%

Unlabeled Real Images: 50
```

```
        |
        v
Classified as Real = 30
        |
        v
Proportion = 60%
```

### 4. Real-world example

A flower recognition model correctly identifies 40 out of 50 labeled flower images. Among another 50 unlabeled real flower images, it identifies 30 as real.

### 5. Advantages

1. Helps measure classification performance.
2. Shows how well the model identifies real images.

### 6. Disadvantages

1. Accuracy alone does not show all classification errors.
2. The proportion does not measure exact class correctness.

### 7. Applications

- Model evaluation.
- Image classification.
- GAN performance measurement.

*Final Answers: 80% accuracy and 60% proportion.*

---

## Q6. Explain the working of a CGAN discriminator and identify the inputs it receives during training.

### 1. Definition

A CGAN discriminator is a neural network that checks whether an image is real or fake and whether it matches the given condition.

### 2. Explanation

The discriminator receives two important inputs:

1. **Image:** A real image or a generated fake image.
2. **Condition:** A class label or other additional information.

Working:

- It receives a real image with its correct condition.
- It receives a generated image with the condition used to generate it.
- It checks whether the image is realistic.
- It also checks whether the image matches the condition.
- Finally, it produces a real or fake decision.

The discriminator tries to correctly identify fake images, while the generator tries to fool it.

## 3. Diagram

```
Real Image + Condition
         |
         |
         v
    Discriminator
         ^
         |
Generated Image + Condition
         |
         v
    Real / Fake
```

## 4. Real-world example

Suppose the condition is digit 5.

- A real image of digit 5 should be accepted.
- A generated image that looks fake should be rejected.
- An image of digit 2 paired with condition 5 should also be treated as an incorrect pair.

## 5. Advantages

1. Checks image quality.
2. Checks whether the output matches the condition.
3. Helps the generator improve.

### 6. Disadvantages

1. Needs correctly labeled training data.
2. Training may become unstable.
3. May fail to check conditions properly if poorly trained.

### 7. Applications

- Text-to-image generation.
- Digit generation.
- Face generation.
- Image translation.

---

## Q7. Explain two benefits and two limitations of Semi-Supervised GANs.

### 1. Definition

A Semi-Supervised GAN is a model that learns using a small amount of labeled data, a large amount of unlabeled data and generated fake samples.

### 2. Explanation

**Two benefits:**

1. **Reduces labeling cost:** It does not require labels for every training image.
2. **Uses unlabeled data:** It learns additional information from unlabeled real images, which can improve classification.

**Two limitations:**

1. **Training complexity:** It contains both a generator and discriminator/classifier, making training difficult.
2. **Training instability:** If the generator produces poor images or the networks become unbalanced, performance may decrease.

### 3. Diagram

```
Labeled Data
     +
```

```
Unlabeled Data
     +
Generated Data
     |
     v
Semi-Supervised GAN
     |
     v
Improved Learning
```

## 4. Real-world example

A hospital has 5,000 medical images, but only 500 images are labeled by doctors. A Semi-Supervised GAN can use the remaining 4,500 unlabeled images to improve learning.

## 5. Advantages

1. Requires fewer labeled images.
2. Uses large unlabeled datasets.
3. Can improve classification performance.

## 6. Disadvantages

1. Difficult to train.
2. Requires careful parameter tuning.
3. Can suffer from mode collapse and instability.

## 7. Applications

- Medical image classification.
- Face recognition.
- Object detection.
- Image recognition.

---

# Q8. Explain the architecture of a CGAN discriminator. How does it evaluate an image together with its condition or class label?

## 1. Definition

The CGAN discriminator is a neural network that receives an image and its condition to determine whether the image is real or fake and whether it matches the condition.

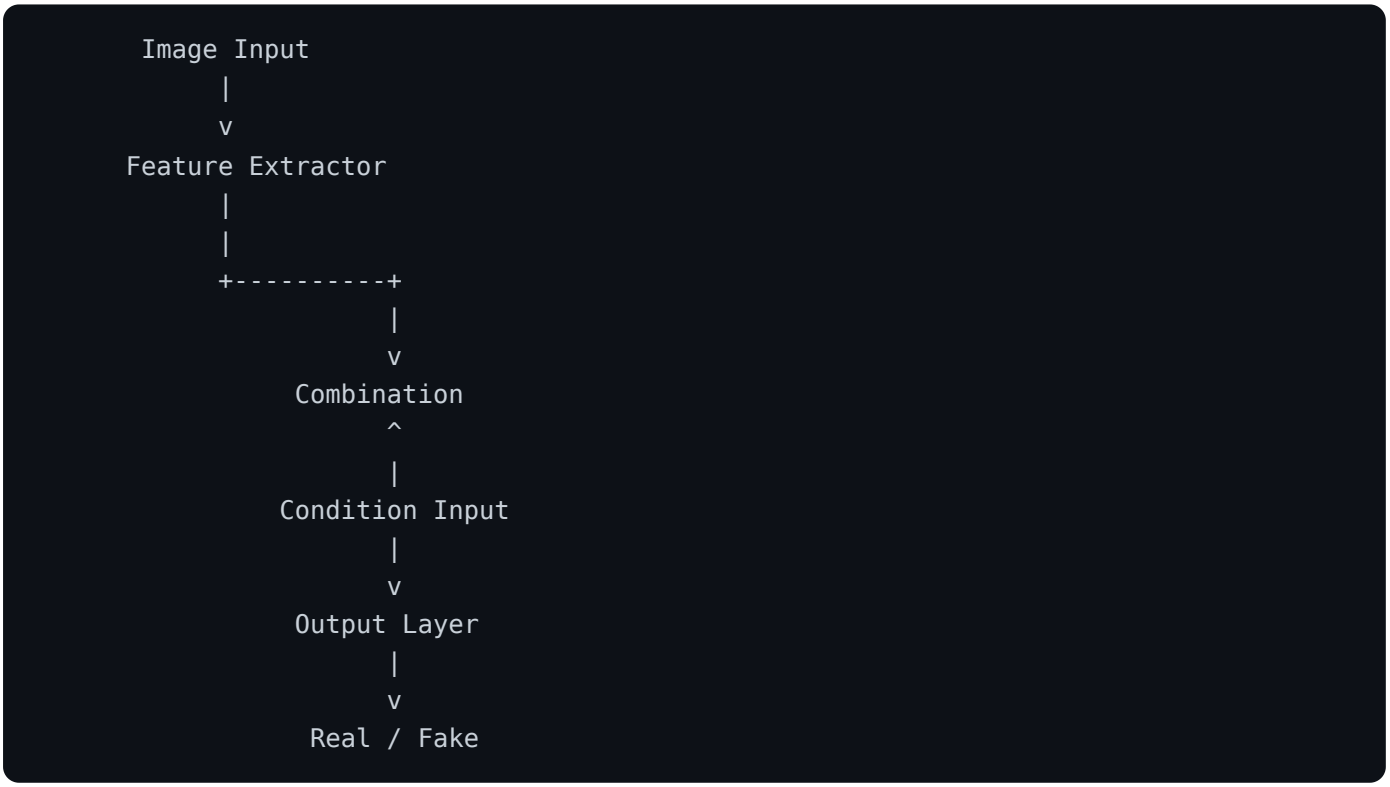
## 2. Explanation

The architecture contains the following parts:

1. **Image Input:** Receives a real or generated image.
2. **Condition Input:** Receives a class label or condition.
3. **Feature Extraction:** Convolutional layers extract important image features.
4. **Combination:** Image features are combined with the condition information.
5. **Output Layer:** Produces a real/fake score.

The discriminator checks both image quality and condition matching.

## 3. Diagram




## 4. Real-world example

Suppose a CGAN generates handwritten digits.

- Image contains digit 7.
- Condition also says 7.
- The discriminator checks whether the image looks real and matches digit 7.

If the image shows digit 2 but the condition says 7, the pair is incorrect.

### 5. Advantages

1. Checks both image quality and class information.
2. Improves controlled image generation.
3. Helps generate condition-matching images.

### 6. Disadvantages

1. Requires correct labels.
2. Architecture is more complex.
3. Incorrect conditions can affect training.

### 7. Applications

- Image generation.
- Handwritten digit recognition.
- Face generation.
- Image translation.

---

## Q9. Describe the complete training procedure of a Conditional GAN, from input preparation to generator and discriminator updates.

### 1. Definition

CGAN training is the process of training the generator and discriminator together to produce realistic images according to the given condition.

### 2. Explanation

The complete training process includes the following steps:

**Step 1: Prepare the dataset**

Collect real images with their corresponding class labels or conditions.

**Step 2: Preprocess the images**

Resize and normalize images before training.

**Step 3: Generate random noise**

Give a random noise vector to the generator.

**Step 4: Provide the condition**

Give a class label or condition along with the noise.

**Step 5: Generate fake images**

The generator produces an image using noise and condition.

**Step 6: Train the discriminator**

The discriminator receives:

- Real images with correct conditions.
- Fake images with their corresponding conditions.

It learns to identify real and fake images.

**Step 7: Train the generator**

The generator receives feedback from the discriminator and tries to generate more realistic images that match the condition.

**Step 8: Repeat training**

Repeat these steps for multiple epochs until the generated images become realistic.

## 3. Diagram






## 4. Real-world example

A clothing company trains a CGAN using images of shirts, pants and shoes.

When the condition is set to shirt, the generator learns to produce realistic shirt images.

## 5. Advantages

1. Generates controlled images.
2. Can generate multiple classes.
3. Improves image generation through feedback.

## 6. Disadvantages

1. Requires labeled training data.
2. Training takes time.
3. Generator and discriminator may become unbalanced.

## 7. Applications

- Fashion image generation.
- Face generation.
- Text-to-image generation.

- Image translation.

---

## Q10. Explain the role of labeled, unlabeled and generated samples in Semi-Supervised GAN training.

### 1. Definition

A Semi-Supervised GAN uses three types of samples: labeled real samples, unlabeled real samples and generated fake samples.

Each sample has a different role in training.

### 2. Explanation

**1. Labeled samples**

- These images contain correct class labels.
- They help the model learn the correct categories.
- Example: Cat, dog and bird images with labels.

**2. Unlabeled samples**

- These are real images without class labels.
- They help the model understand the real data distribution.
- They are classified into one of the K real classes rather than the fake class.

**3. Generated samples**

- These are created by the generator.
- They help the discriminator learn to identify fake images.
- They belong to the additional K+1th class.

### 3. Diagram

```
Labeled Real Images
        |
        v
  Correct K Classes

Unlabeled Real Images
        |
        v
```

```
   Real K Classes

Generated Fake Images
        |
        v
    Fake K+1 Class
```

## 4. Real-world example

In a fruit classification system:

- Labeled images: Apple and banana images with labels.
- Unlabeled images: Real fruit images without labels.
- Generated images: Artificially created fruit images.

All three help the model learn.

## 5. Advantages

1. Uses different types of training data.
2. Reduces manual labeling.
3. Improves learning.

## 6. Disadvantages

1. Training is complex.
2. Poor fake images may affect learning.
3. Incorrect predictions may reduce accuracy.

## 7. Applications

- Fruit classification.
- Medical image analysis.
- Object recognition.
- Face recognition.

---

# Q11. Explain how class labels can guide image generation in a CGAN, using a suitable example.

## 1. Definition

In a CGAN, class labels are used as conditions to guide the generator to produce images belonging to a particular class.

## 2. Explanation

1. First, class labels are represented as numerical values or one-hot vectors.
2. Random noise is generated.
3. The noise and class label are combined.
4. The combined input is given to the generator.
5. The generator creates an image of the required class.
6. The discriminator checks whether the image is real and matches the label.

Formula:

$$G(z, y) = \text{Generated Image}$$

Where:

- z = Random noise.
- y = Class label.

## 3. Diagram

```
Random Noise
     +
Class Label (7)
     |
     v
  Generator
     |
     v
Image of Digit 7
     |
     v
Discriminator
     |
     v
Real / Fake
```

### 4. Real-world example

Consider the MNIST handwritten digit dataset.

- Label 5 generates digit 5.
- Label 7 generates digit 7.
- Label 9 generates digit 9.

Thus, class labels control what type of image is generated.

### 5. Advantages

1. Generates a selected class.
2. Provides better control.
3. Creates different variations of the same class.

### 6. Disadvantages

1. Needs correct labels.
2. Wrong labels produce incorrect outputs.
3. Generator may not follow the condition perfectly.

### 7. Applications

- Handwritten digit generation.
- Face generation.
- Fashion image generation.
- Object image synthesis.

---

## Q12. Compare a Semi-Supervised GAN with a fully supervised classifier in terms of labeled data, architecture, training objectives and limitations.

### 1. Definition

A fully supervised classifier learns using labeled data only, whereas a Semi-Supervised GAN learns using labeled data, unlabeled real data and generated fake samples.

## 2. Explanation

The main differences are:

## 3. Difference table

| Feature | Fully Supervised Classifier | Semi-Supervised GAN |
|---|---|---|
| 1. Data | Uses labeled data | Uses labeled and unlabeled data |
| 2. Labels | Needs labels for training examples | Needs labels for only some real examples |
| 3. Architecture | Mainly classifier | Generator and discriminator/classifier |
| 4. Training | Uses supervised loss | Uses supervised and unsupervised losses |
| 5. Fake images | Not required | Generated by generator |
| 6. Output | Predicts real classes | Predicts K real classes and one fake class |
| 7. Complexity | Simple | More complex |
| 8. Training | Easier | More difficult |
| 9. Main advantage | Easy to train | Uses unlabeled data |
| 10. Limitation | Requires many labels | May have training instability |

## 4. Diagram

### Fully Supervised Classifier

```
Labeled Images
      |
      v
  Classifier
      |
      v
 Predicted Class
```

### Semi-Supervised GAN

```
 Labeled + Unlabeled Images
            |
            v
       Discriminator
            ^
            |
```

```
        Generated Images
              ^
              |
          Generator
```

### 5. Real-world example

Suppose we want to classify 10,000 flower images.

- Fully supervised classifier: Requires labels for all training images.
- Semi-Supervised GAN: Can use 1,000 labeled images and 9,000 unlabeled images.

### 6. Advantages

1. Semi-Supervised GAN reduces labeling effort.
2. Fully supervised classification is easier to implement.
3. Semi-Supervised GAN can learn from additional unlabeled data.

### 7. Disadvantages

1. Fully supervised models require more labeled data.
2. Semi-Supervised GANs are more complex and difficult to train.

### 8. Applications

- Image classification.
- Medical image analysis.
- Face recognition.
- Object recognition.

---

## Q13. Outline the major implementation steps required to build and train a CGAN.

### 1. Definition

CGAN implementation is the process of building a generator and discriminator that use conditions to generate realistic images.

## 2. Explanation

The major implementation steps are:

**Step 1: Collect Dataset**

Collect images with their corresponding labels.

**Step 2: Preprocess Data**

Resize images, normalize pixel values and prepare labels.

**Step 3: Encode Conditions**

Convert class labels into one-hot vectors or suitable numerical representations.

**Step 4: Build Generator**

Create a generator that accepts random noise and condition.

**Step 5: Build Discriminator**

Create a discriminator that accepts an image and its condition.

**Step 6: Select Loss Function**

Use suitable generator and discriminator loss functions.

**Step 7: Train Discriminator**

Train it using real and generated images.

**Step 8: Train Generator**

Train it to generate realistic images that match the conditions.

**Step 9: Evaluate Results**

Check image quality, diversity and whether the generated images match the requested conditions.

## 3. Diagram



```
          |
          v
   Build Generator
          |
          v
 Build Discriminator
          |
          v
    Train Both Networks
          |
          v
    Evaluate Images
          |
          v
      Final CGAN
```

## 4. Real-world example

To build a CGAN for handwritten digits:

1. Collect MNIST dataset.
2. Prepare images and labels.
3. Build generator and discriminator.
4. Train using noise and digit labels.
5. Generate digits from 0 to 9.

## 5. Advantages

1. Generates class-specific images.
2. Provides controlled image generation.
3. Can generate multiple image categories.

## 6. Disadvantages

1. Requires labeled data.
2. Needs high computational resources.
3. Training may be unstable.

## 7. Applications

- Image generation.
- Digit generation.
- Face generation.

- Fashion design.

---

## Q14. Describe the architecture of a Semi-Supervised GAN and explain the purpose of the additional (K+1)th class.

### 1. Definition

A Semi-Supervised GAN consists of a generator and a discriminator/classifier. The discriminator has K+1 output classes.

Here:

- K = Number of real classes.
- K+1 = Additional fake class.

### 2. Explanation

The architecture works as follows:

1. The generator receives random noise and creates fake images.
2. The discriminator receives labeled real images.
3. It classifies them into their correct K real classes.
4. It also receives unlabeled real images and learns to classify them as real.
5. Generated images are assigned to the additional K+1th class.

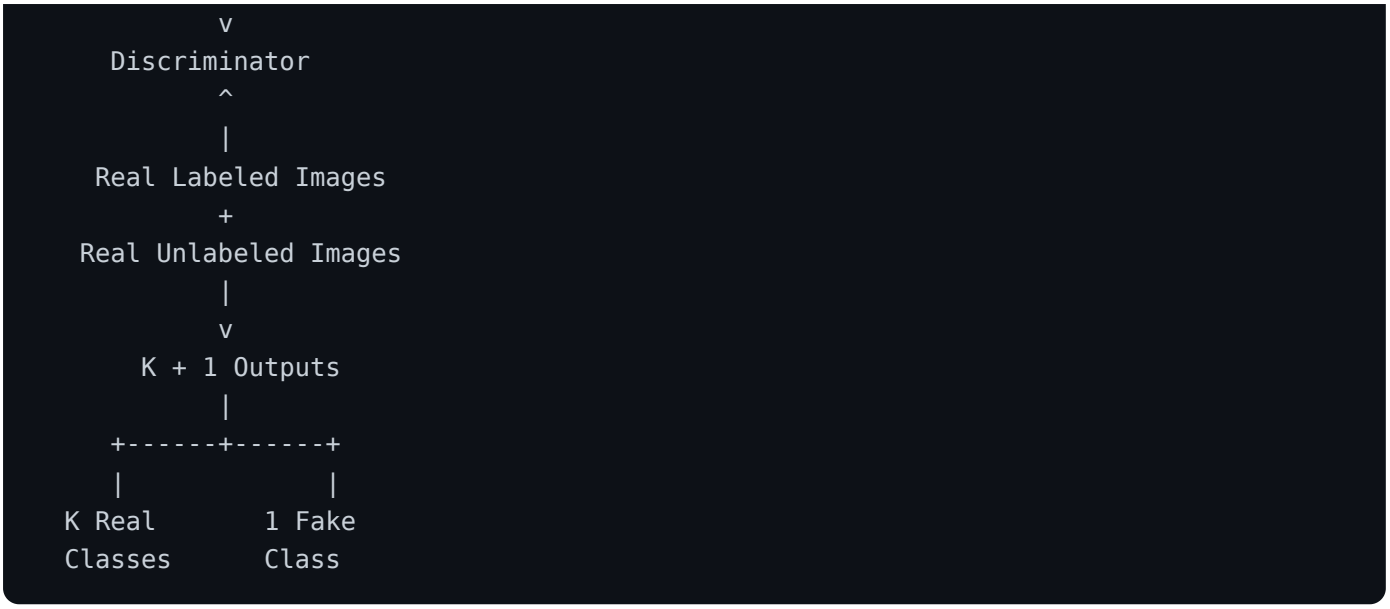
**Purpose of the K+1th class:**

The extra class helps the discriminator identify generated fake images separately from real images.

For example, if there are 10 real classes, the discriminator has 11 output classes.

### 3. Diagram






## 4. Real-world example

In the MNIST dataset:

- Class 0 = Digit 0.
- Class 1 = Digit 1.
- ...
- Class 9 = Digit 9.
- Class 10 = Generated fake images.

Therefore, the discriminator has 11 output classes.

## 5. Advantages

1. Learns real and fake differences.
2. Uses unlabeled real data.
3. Helps improve classification.

## 6. Disadvantages

1. More complex architecture.
2. Requires careful training.
3. Poor fake images may affect performance.

## 7. Applications

- Handwritten digit classification.
- Image recognition.

- Medical image analysis.
- Semi-supervised learning.

---

## Q15. What problems may occur when the condition supplied to a CGAN is incorrect or inconsistent with the training image?

### 1. Definition

A condition is additional information given to a CGAN, such as a class label or text description. If the condition does not match the actual image, it creates incorrect training information.

### 2. Explanation

The following problems may occur:

1. **Wrong image generation:** The generator may produce an image that does not match the requested class.
2. **Poor image quality:** Incorrect conditions can confuse the generator and reduce image quality.
3. **Training difficulty:** The discriminator receives inconsistent image-condition pairs and learns incorrect relationships.
4. **Mixed features:** Generated images may contain features from different classes.
5. **Reduced accuracy:** The model may fail to generate the correct output.

To avoid these problems, training images must have correct labels and conditions.

### 3. Diagram

```
    Real Image
       +
 Incorrect Label
       |
       v
 Wrong Training Input
       |
       v
     CGAN
       |
       v
Incorrect Generated
      Image
```

### 4. Real-world example

Suppose an image contains a cat, but its condition is labeled as dog.

The CGAN receives incorrect information and may learn the wrong relationship between images and labels.

### 5. Advantages

1. Correct conditions improve output quality.
2. Proper labeling improves training.
3. Consistent data improves image generation.

### 6. Disadvantages

1. Wrong conditions produce incorrect images.
2. Training becomes difficult.
3. Output quality and accuracy decrease.

### 7. Applications

- Text-to-image generation.
- Image classification.
- Face generation.
- Image translation.

---

## Q16. Explain the training objectives of a Semi-Supervised GAN. Differentiate between supervised classification loss and unsupervised loss.

### 1. Definition

The training objective of a Semi-Supervised GAN is to classify labeled real images correctly, identify unlabeled real images as real, recognize generated images as fake and improve the generator.

### 2. Explanation

There are three main objectives:

**1. Supervised Classification Loss**

- Uses labeled real images.
- Compares predicted classes with actual labels.
- Helps the model classify images correctly.

Formula:

$$L_{sup} = -E_{(x,y)}[\log p(y|x)]$$

**2. Unsupervised Loss**

- Uses unlabeled real images and generated fake images.
- Encourages the model to classify real images as real.
- Encourages the model to classify fake images as fake.

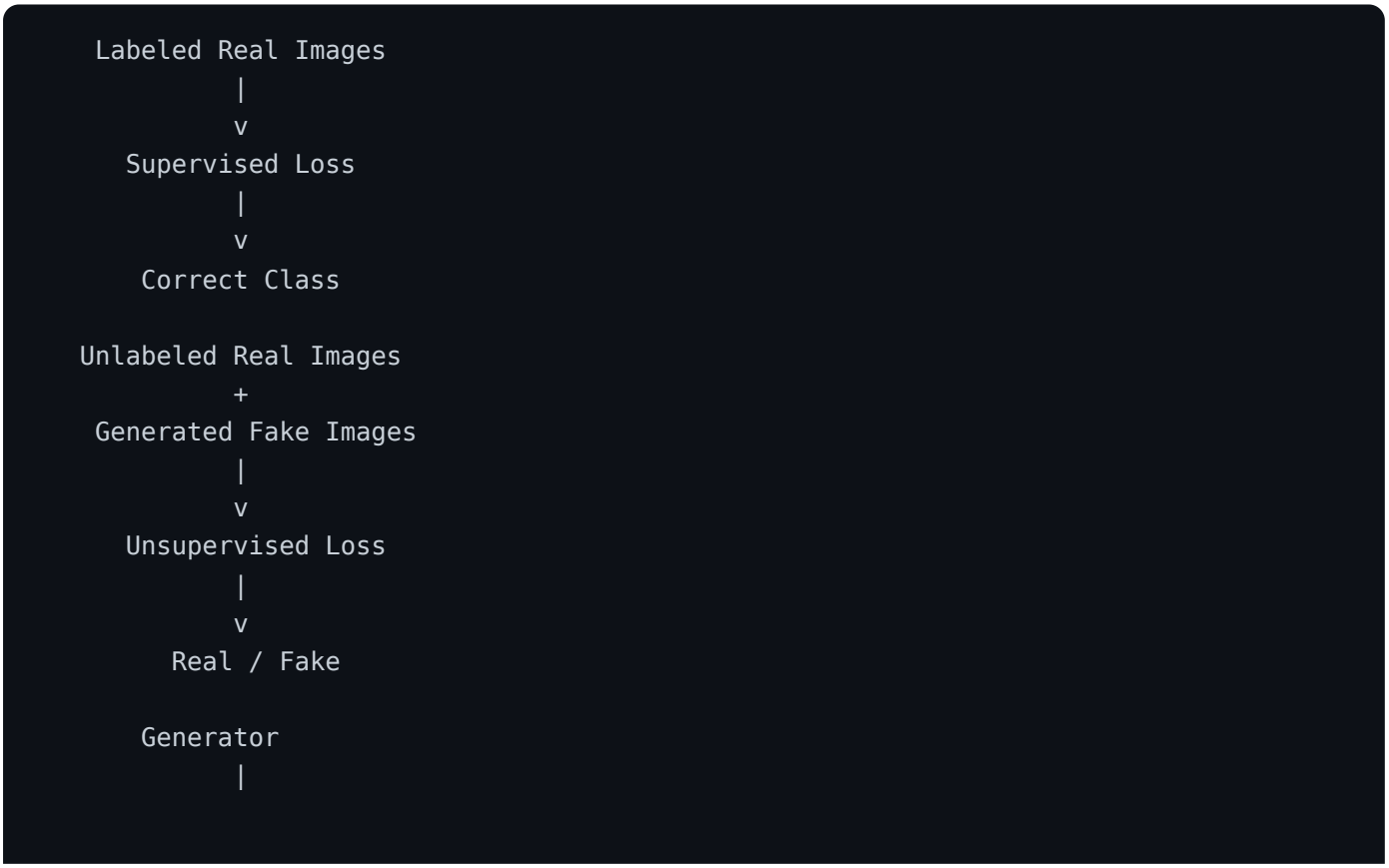
Formula:

$$L_{unsup} = -E_x[\log(1 - p(fake|x))]$$
$$-E_z[\log p(fake|G(z))]$$

**3. Generator Objective**

- The generator tries to create realistic images.
- It tries to fool the discriminator into classifying generated images as real.

## 3. Diagram



```
            v
    Realistic Fake Images
```

## 4. Difference table

| Supervised Loss | Unsupervised Loss |
|---|---|
| Uses labeled real images | Uses unlabeled real and generated images |
| Requires class labels | Does not require exact class labels for real images |
| Predicts correct class | Separates real and fake images |
| Uses classification loss | Uses real/fake loss |
| Learns class information | Learns data distribution and real/fake distinction |

## 5. Real-world example

A fruit recognition system uses labeled apple and banana images for supervised loss and unlabeled fruit images along with generated fake images for unsupervised loss.

## 6. Advantages

1. Uses both labeled and unlabeled data.
2. Improves classification learning.
3. Reduces labeling requirements.

## 7. Disadvantages

1. Training is complex.
2. Loss balancing is difficult.
3. Poor generated samples may affect learning.

## 8. Applications

- Image classification.
- Medical image analysis.
- Object recognition.
- Face recognition.

---

# Q17. Compare an unconditional GAN with a Conditional GAN in terms of input, control over output and applications.

## 1. Definition

**Unconditional GAN:** A GAN that generates images using only random noise.

**Conditional GAN:** A GAN that generates images using random noise along with additional information called a condition.

## 2. Explanation

An unconditional GAN generates images without allowing the user to directly select the output class.

A Conditional GAN provides control over the output using class labels, text or other conditions.

## 3. Difference table

| Feature | Unconditional GAN | Conditional GAN |
|---|---|---|
| 1. Input | Random noise | Noise + condition |
| 2. Control | No direct control | Controlled output |
| 3. Labels | Not required as generator input | Condition information required |
| 4. Generator | G(z) | G(z,y) |
| 5. Discriminator | Receives image | Receives image + condition |
| 6. Output | Random type of image | Required type of image |
| 7. Complexity | Simple | More complex |
| 8. Training | Comparatively simple | Requires condition-based training |
| 9. Example | Random animal images | Selected dog or cat image |
| 10. Applications | General image generation | Controlled image generation |

## 4. Diagram

**Unconditional GAN**

```
Random Noise
     |
```

```
      v
 Generator
      |
      v
Generated Image
```

**Conditional GAN**

```
Random Noise + Condition
         |
         v
      Generator
         |
         v
 Required Generated Image
```

## 5. Real-world example

Suppose we are generating animal images.

- Unconditional GAN: Generates any animal randomly.
- Conditional GAN: Generates a dog when the condition is dog and a cat when the condition is cat.

## 6. Advantages

**Unconditional GAN:**

1. Simple architecture.
2. Does not require condition input.

**Conditional GAN:**

1. Provides control over output.
2. Generates specific classes.

## 7. Disadvantages

**Unconditional GAN:**

1. Cannot directly select the output class.
2. Output may be unpredictable.

**Conditional GAN:**

1. Requires condition information.

2. More difficult to train.

### 8. Applications

- Unconditional GAN: General image synthesis and creative image generation.
- Conditional GAN: Face generation, digit generation, text-to-image generation and image translation.

---

## Q18. Explain how a Semi-Supervised GAN can learn from unlabeled real images in addition to labeled images.

### 1. Definition

A Semi-Supervised GAN learns from unlabeled real images by using them as additional training data, even when their exact class labels are not available.

### 2. Explanation

1. The model receives a small number of labeled images.
2. It learns the correct classes from labeled images.
3. It also receives a large number of unlabeled real images.
4. The discriminator learns that these images belong to real data and should not be classified as fake.
5. The generator creates fake images.
6. The discriminator learns to separate generated images from real images.
7. By using both labeled and unlabeled data, the model improves its learning.

### 3. Diagram



```
        ^
        |
    Generator
        |
        v
Better Learning
```

## 4. Real-world example

Suppose we have 5,500 fruit images.

- 500 images are labeled.
- 5,000 images are unlabeled.

The model learns the correct fruit classes from the 500 labeled images and uses the remaining unlabeled images to learn more about real fruit images.

## 5. Advantages

1. Uses large amounts of unlabeled data.
2. Reduces manual labeling cost.
3. Can improve classification performance.

## 6. Disadvantages

1. Training is complex.
2. Incorrect predictions can affect learning.
3. Unlabeled data must be relevant to the task.

## 7. Applications

- Fruit classification.
- Medical image analysis.
- Object recognition.
- Face recognition.

---

# Q19. Explain the implementation steps of a Semi-Supervised GAN and discuss any two challenges encountered during training.

## 1. Definition

A Semi-Supervised GAN is implemented by building a generator and a K+1 class discriminator/classifier and training them using labeled, unlabeled and generated data.

## 2. Explanation

**Implementation steps:**

**Step 1: Collect Dataset**

Collect a small labeled dataset and a large unlabeled dataset.

**Step 2: Preprocess Data**

Resize and normalize the images.

**Step 3: Build Generator**

Create a generator that accepts random noise and produces fake images.

**Step 4: Build Discriminator**

Create a discriminator with K+1 output classes.

**Step 5: Train with Labeled Data**

Use supervised loss to classify labeled images correctly.

**Step 6: Train with Unlabeled Data**

Use unsupervised loss to identify unlabeled real images as real.

**Step 7: Train with Generated Data**

Train the discriminator to classify generated images as fake.

**Step 8: Update Generator**

The generator tries to produce realistic images that can fool the discriminator.

**Step 9: Evaluate the Model**

Check classification accuracy and generated image quality.

## 3. Diagram




## 4. Real-world example

A flower classification system uses 1,000 labeled flower images and 5,000 unlabeled flower images. A generator creates fake flower images to support the training process.

## 5. Advantages

1. Uses both labeled and unlabeled data.
2. Reduces labeling effort.
3. Improves learning from limited labeled examples.

## 6. Disadvantages and challenges

### Challenge 1: Training instability

- Generator and discriminator may become unbalanced.
- One network may learn faster than the other.
- This affects the overall training process.

### Challenge 2: Poor generated images

- Initially, the generator may produce unrealistic images.

- These images may be too easy to identify as fake.
- This can reduce the usefulness of training.

Other challenges include class imbalance and selecting suitable loss weights.

### 7. Applications

- Medical image classification.
- Object recognition.
- Face recognition.
- Image classification.

---

## Q20. Explain how generated samples contribute to the training of a Semi-Supervised GAN.

### 1. Definition

Generated samples are artificial or fake images created by the generator using random noise. They help the discriminator learn the difference between real and fake images.

### 2. Explanation

1. The generator receives random noise.
2. It creates fake images.
3. These generated images are given to the discriminator.
4. The discriminator learns to classify them into the additional K+1th fake class.
5. The generator receives feedback from the discriminator.
6. It tries to produce more realistic images.
7. This process continues until the generated images become more realistic.

**Main purpose:** Generated samples help the discriminator identify fake images and help the generator improve image quality.

### 3. Diagram

```
      Random Noise
           |
           v
       Generator
```

```
            |
            v
      Generated Images
            |
            v
      Discriminator
            |
            v
       Fake Class
        (K+1)
            |
            v
      Feedback to G
            |
            v
    Better Fake Images
```

## 4. Real-world example

Suppose we are training a handwritten digit recognition system.

- The generator creates artificial handwritten digits.
- The discriminator learns to identify these as fake.
- The generator improves its images to make them look like real handwritten digits.

Thus, generated samples help both networks improve.

## 5. Advantages

1. Provides fake examples for training.
2. Helps distinguish real and fake images.
3. Improves generator performance.
4. Reduces dependence on labeled data.

## 6. Disadvantages

1. Poor generated samples may not provide useful learning.
2. Generator may produce similar images repeatedly (mode collapse).
3. Training can become unstable.

## 7. Applications

- Image classification.
- Handwritten digit generation.

- Synthetic image generation.
- Object recognition.

---

# PART B: QUICK REVISION

These are the most important points to remember before your examination.

## 1. Important formulas

| Topic | Formula |
|---|---|
| CGAN Generator | $G(z, y)$ |
| Supervised Loss | $L_{sup} = -E[\log p(y \mid x)]$ |
| Unsupervised Loss | $L_{unsup} = -E_x[\log(1 - p(fake \mid x))] - E_z[\log p(fake \mid G(z))]$ |
| Classification Accuracy | $\frac{Correct}{Total} \times 100$ |
| Proportion | $\frac{Number\ of\ selected\ samples}{Total}$ |

## 2. Important concepts

| Concept | Remember this |
|---|---|
| CGAN | Generates images based on conditions |
| Noise vector | Provides randomness |
| Condition | Controls the generated output |
| Generator | Creates fake images |
| Discriminator | Identifies real and fake images |
| Labeled data | Contains correct class labels |
| Unlabeled data | Does not contain class labels |
| Generated data | Artificial images created by generator |

| Concept | Remember this |
| --- | --- |
| K+1 class | Extra class for fake images |
| Supervised loss | Learns correct classes |
| Unsupervised loss | Separates real and fake images |

## 3. Most important diagrams to practise

### 1. CGAN Generator

```
Noise + Condition
       |
       v
    Generator
       |
       v
  Generated Image
```

### 2. CGAN Discriminator

```
Image + Condition
       |
       v
  Discriminator
       |
       v
    Real / Fake
```

### 3. Semi-Supervised GAN

```
Labeled Real Images
Unlabeled Real Images
Generated Fake Images
        |
        v
   Discriminator
        |
        v
    K + 1 Classes
```

### 4. CGAN Training

```
Noise + Condition
       |
       v
    Generator
       |
       v
    Fake Image
       |
       v
```

```
Discriminator
     |
     v
  Feedback
     |
     v
Update Networks
```

## 4. Important questions for revision

Give extra attention to these topics because they cover the main concepts of Module 4:

1. Q2 – Training process of Semi-Supervised GAN.
2. Q3 – CGAN generator architecture.
3. Q4 – Supervised and unsupervised loss.
4. Q5 – Numerical problem.
5. Q8 – CGAN discriminator architecture.
6. Q9 – Complete CGAN training procedure.
7. Q12 – Semi-Supervised GAN vs fully supervised classifier.
8. Q14 – K+1 class architecture.
9. Q16 – Training objectives.
10. Q17 – Unconditional GAN vs Conditional GAN.

**Exam preparation tip:** For a 5-mark answer, write the definition, explanation, diagram and example. Add advantages, disadvantages and applications when space permits or when the question asks for them. For comparison questions, use tables. For numerical questions, always show the formula, substitution and final answer.